

LC-3 Instruction Encodings

Arith./Logic		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	ADD†	0	0	0	1	DR		SR1		0			0	0	SR2									
	ADD†	0	0	0	1	DR		SR1		1	imm5													
	AND†	0	1	0	1	DR		SR1		0			0	0	SR2									
	AND†	0	1	0	1	DR		SR1		1	imm5													
	NOT†	1	0	0	1	DR		SR		1			1	1	1	1	1							
Load/Store		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	LD†	0	0	1	0	DR		PCoffset9																
	ST	0	0	1	1	SR		PCoffset9																
	LDI†	1	0	1	0	DR		PCoffset9																
	STI	1	0	1	1	SR		PCoffset9																
	LDR†	0	1	1	0	DR		BaseR		offset6														
	STR	0	1	1	1	SR		BaseR		offset6														
Branch		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	JMP	1	1	0	0	0		0	0	BaseR		0						0	0	0	0			
	BR	0				0	0	0	cn	cz	cp	PCoffset9												
	JSRR	0	1	0	0	0		0	0	BaseR		0						0	0	0	0	0		
	JSR	0				1	0	0	1	PCoffset11														
Misc.		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0							
	LEA	1	1	1	0	DR		PCoffset9																
	TRAP	1	1	1	1	0			0	0	0	trapvect8												
	RTI	1	0	0	0	0												0	0	0	0	0	0	
	illegal	1	1	0	1	x												x	x	x	x	x	x	x

[†] Updates condition code (CC)

LC-3 MMIO Registers

		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFE00	KBSR	Rdy	(undefined)														
0xFE02	KBDR	0	0	0	0	0	0	0	0	ASCII code							
0xFE04	DSR	Rdy	(undefined)														
0xFE06	DDR	0	0	0	0	0	0	0	0	ASCII code							
0xFFFC	PSR	Priv	(undefined)				PRIO				(undefined)				N	Z	P

LC-3 Trap Vectors

- 0x20 → GETC
- 0x21 → PUTC
- 0x22 → PUTS
- 0x25 → HALT

LC-3 Memory Map

0x0000	Trap vector table
0x00FF	
0x0100	Interrupt vector table
0x01FF	
0x0200	OS memory ↑ Supervisor Stack
0x2FFF	
0x3000	Code User memory ↑ User stack
0xFDFF	MMIO
0xFE00	
0xFFFF	

LC-3 Datapath Control Signals

(Use these for filling out LC-3 FSM *state transition diagrams*)

Signals to assert:

- LD.MAR
- LD.MDR
- LD.IR
- LD.REG
- LD.CC
- LD.DDR
- GatePC
- GateMDR
- GateALU
- GateMARMUX
- GateSP
- GatePSR
- MIO.EN

Signals with named values:

- PCMUX: PC+1, BUS, ADDER
- SR1MUX: IR[11:9], IR[8:6]
- DRMUX: IR[11:9], 7
- ALUK: ADD, AND, NOTA, PASSA
- ADDR1MUX: PC, BaseR
- ADDR2MUX: ZERO, offset6, PCoffset9, PCoffset11
- MARMUX: IR[7:0], ADDER
- INMUX: MEM, KBSR, KBDR, DSR
- SPMUX: Saved.USP, Saved.SSP
- MIO.RW: R, W

LC-3 Microcode Control Signal Bit Assignments

(Use these for filling out bits in *microcode tables*)

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|--|---|
| <ul style="list-style-type: none"> • PCMUX: <ul style="list-style-type: none"> • 00₂ → PC+1 • 01₂ → BUS • 10₂ → ADDER • SR1MUX: <ul style="list-style-type: none"> • 00₂ → IR[11:9] • 01₂ → IR[8:6] • 10₂ → 6 • DRMUX: <ul style="list-style-type: none"> • 00₂ → IR[11:9] • 01₂ → 7 • 10₂ → 6 • ALUK: <ul style="list-style-type: none"> • 00₂ → ADD • 01₂ → AND • 10₂ → NOTA • 11₂ → PASSA • SPMUX: <ul style="list-style-type: none"> • 0 → Saved.USP • 1 → Saved.SSP | <ul style="list-style-type: none"> • ADDR1MUX: <ul style="list-style-type: none"> • 0 → PC • 1 → BaseR • ADDR2MUX: <ul style="list-style-type: none"> • 00₂ → ZERO • 01₂ → offset6 • 10₂ → PCoffset9 • 11₂ → PCoffset11 • MARMUX: <ul style="list-style-type: none"> • 0 → IR[7:0] • 1 → ADDER • INMUX: <ul style="list-style-type: none"> • 00₂ → MEM • 01₂ → KBSR • 10₂ → KBDR • 11₂ → DSR • MIO.RW: <ul style="list-style-type: none"> • 0 → R • 1 → W |
|--|---|

